

This is a classic libadwaita "layout-over-styling" conflict. Looking at your `windowcontrols_ovals_review_2026-04-11.md` and the screenshot, the issue isn't that your CSS isn't applying—it clearly is, given the colors—but that Nautilus uses a different headerbar implementation than Settings, which is triggering internal minimum widths you aren't seeing in the inspector.

1. The "Why": Nautilus vs. Settings

The reason it works in **Settings** but fails in **Nautilus** is likely due to the `AdwWindow` vs. `AdwApplicationWindow` implementation. Nautilus often utilizes a `GtkHeaderBar` or an older `AdwHeaderBar` configuration that reserves extra horizontal space for the window control box to prevent "jitter" during window resizing. Because GTK4 CSS lacks `max-width`, you can set a floor (`$24\text{px}`), but the container is forcing a wider ceiling.

2. The Solution: Targeting the Container

To achieve perfect circles, you must shrink the **container** (the `windowcontrols` box itself) while forcing the button to ignore the inherited horizontal expansion.

Try this specific block in your `~/.config/gtk-4.0/gtk.css`:

```
CSS
/* 1. Force the container to be as tight as possible */
windowcontrols {
  border-spacing: 0;
  margin: 0;
  padding: 0;
}

/* 2. Target the button with zeroed-out padding and fixed min-bounds */
windowcontrols button {
  padding: 0 !important;
  margin: 4px 2px; /* Slight horizontal margin to prevent touching */
  min-width: 24px !important;
  min-height: 24px !important;
  border-radius: 100%;
  /* Ensure the button doesn't try to fill the headerbar height */
  vertical-align: center;
  -gtk-icon-size: 14px;
}

/* 3. The Secret Sauce: Force the IMAGE to be the layout master */
windowcontrols button > image {
  padding: 5px !important; /* This creates the 'internal' space for the circle */
  margin: 0 !important;
```

```
min-width: 14px !important;  
min-height: 14px !important;  
}
```

3. Addressing Your Specific Questions

- **Specificity:** windowcontrols button is technically less specific than libadwaita's windowcontrols > button. Use the > child combinator in your rules to match their specificity, or stick with !important for this specific edge case.
- **Symbolic Icons:** They are indeed "fixed" at 16px internally. If you set a button to 24px and the icon is 16px , you only have 4px of padding on each side. If Nautilus's layout engine adds even 1px of internal padding, it becomes an oval.
- **aspect-ratio:** Unfortunately, GTK4 CSS **does not** support the standard web aspect-ratio property.
- **The .text-button theory:** It is unlikely Nautilus is using .text-button for window controls, but it *is* likely using headerbar.default-decoration, which has higher priority. Ensure your selectors start with headerbar windowcontrols button to win the specificity war.

4. Direct Inspector Tip

Since you have the Inspector enabled, don't just look at "Properties." Go to the **Layout** tab for the GtkButton inside the GtkWindowControls. Look for "**Horizontal Alignment**". If it says FILL, that is your culprit. You can't change that via CSS, but you can counteract it by setting margin-left and margin-right to larger values to "squeeze" the button back into a square.

How does the window control rendering look in Nautilus if you temporarily switch to the default Adwaita theme—does the ovaling persist there too?